

Wild Realms - An Interactive Wildlife Education Simulation

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In 4 weeks, our group will demonstrate a simple working prototype of **Wild Realms** that will allow the user to navigate through biomes (aquatic, desert, forest, grassland, and tundra) using a web-based interface. The user will be able to select a species within the biome. The application will display the species info (diet, conservation status, and role in the ecosystem). The user will also be able to select circumstances and watch a simulation of the effects of that circumstance on a selected species.

The actors for this application are the student/learner that explores biomes and interacts with species and the educator that integrates the prototype into a learning activity.

Scenario “Wild Realms”

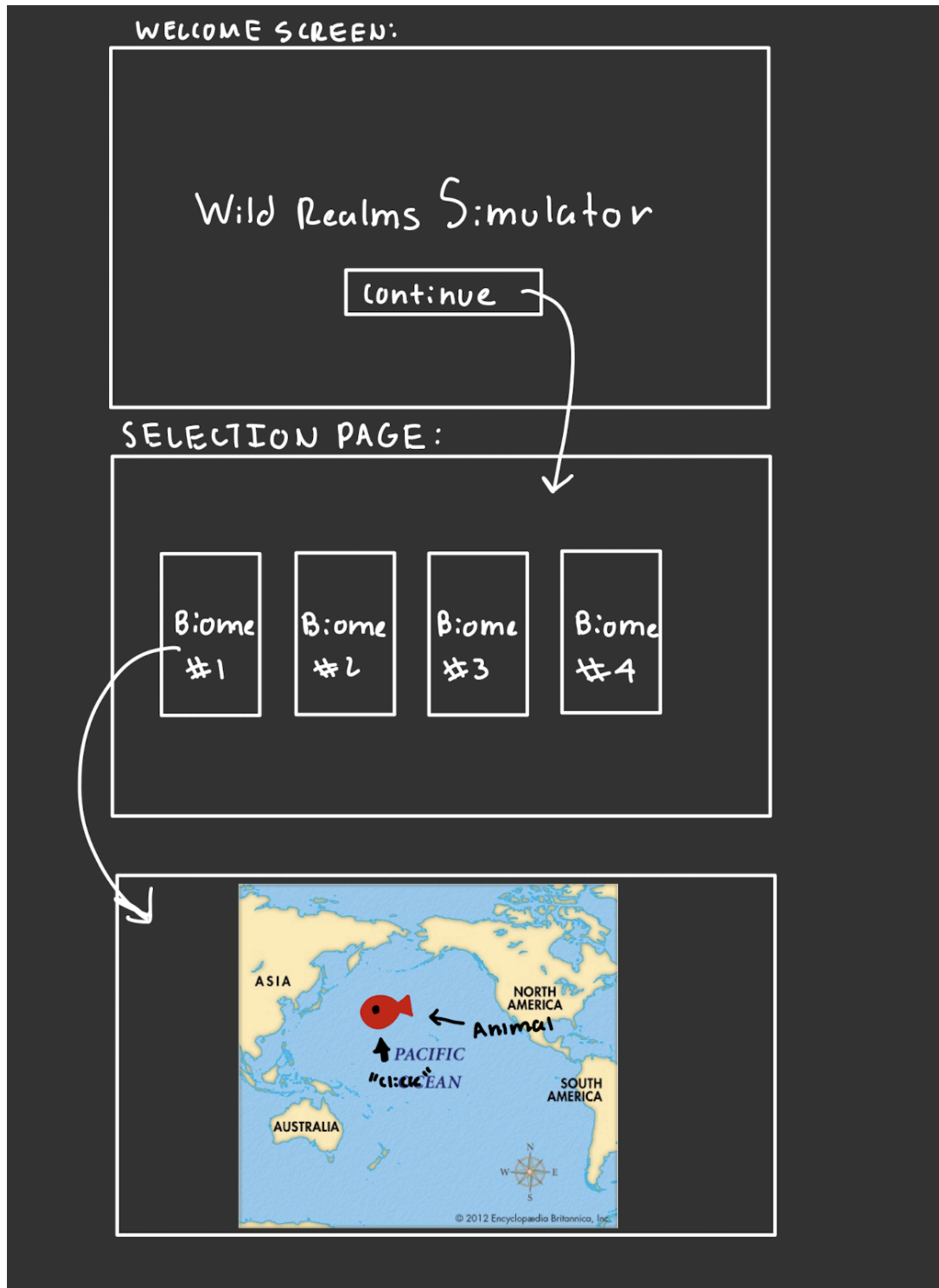
The user will enter the simulation’s URL into their browser’s address bar, which will take them to the main page of the application. The welcome page will load and display the name of the simulator and a “Continue” button. If the user selects the continue button, it will take them to the simulator’s selection page.

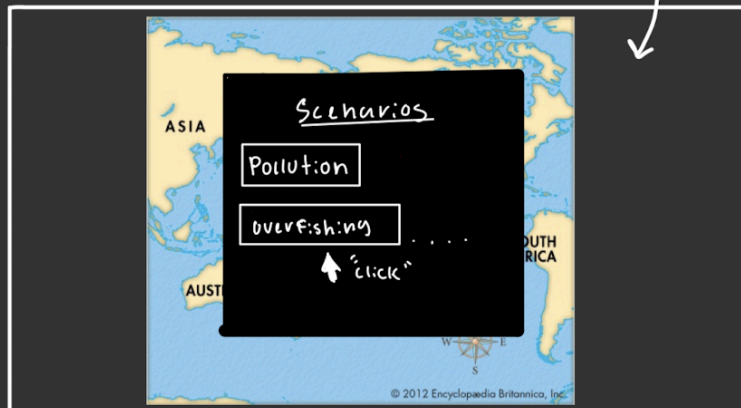
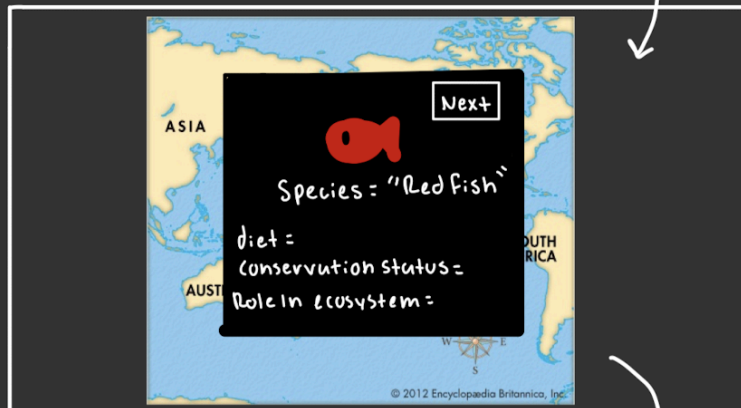
On the selection page, the user will see options to choose a biome (aquatic, desert, forest, grassland, and tundra). Our group plans to code at least three biomes to start, with all five included if time permits. Once the user selects a biome, the simulator will display a globe centered on that biome. From there, the user will see 2D images of animals native to the biome.

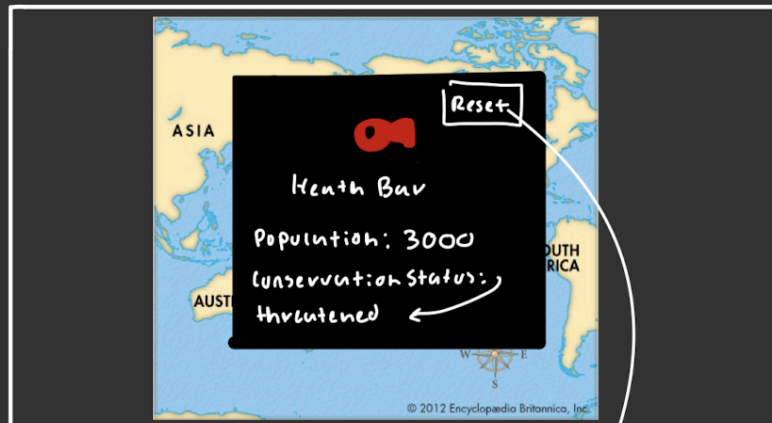
When the user selects an animal image, a pop-up will display information about the species, including its diet, conservation status, and role in the ecosystem. On the same page, a “Next” button will allow the user to move forward. Clicking “Next” takes the user to a page where they can select different circumstances, including: Pollution, Deforestation (Habitat Loss), Climate Change Effects, Overfishing/Hunting/Poaching, Invasive Species, and Conservation Actions (Tree Planting, Pollution Cleanup, Protected Areas). Again, our group plans to aim for 3 circumstances but will code all if time permits.

When the user selects a circumstance, the simulator will show how the attributes for that animal change. Each animal will have a health bar/status meter, a population index (e.g., 100 → 70), and a conservation status label (e.g., “Near Threatened” → “Endangered”). There will be a series of backend calculations that the simulation does at this point to figure out the status of the species using real-world data.

For each circumstance, the simulator will model the effects on the animal over a 12-month period. After the user views the 12-month results, a reset button will allow them to return to the biome selection page and restart the simulation.







SELECTION PAGE:

